

THE FUTURE IS HERE
A technician using a high-tech analyser, capable of doing 2,200 tests per hour, at a Dr Dangs Lab facility in Delhi

close to 75 per cent of our health infrastructure is concentrated in metro cities, 55 million citizens are pushed into poverty each year due to spending on healthcare, and every fourth Indian is at risk of dying of a non-communicable disease before the age of 70. It's in such a scenario that AI-led diagnostics promise to offer affordable options, and ease the burden on scarce infrastructure.

Since AI-based predictive analysis is most useful in the field of diagnostics, the growth in that industry is indicative of where India is headed, even though the numbers may appear modest at the moment. A 2023 report on AI in diagnostics in India by market analysis firm Insights10 has projected its growth from $0.02 billion (Rs 165 crore) in 2022 to $0.24 billion (Rs 1,987 crore) by 2030, registering a CAGR, or compound annual growth rate, of 38 per cent in the period. The Indian medical diagnostics industry as a whole is expected to touch $20 billion (Rs 1.66 lakh crore) by 2026 from $10 billion (Rs 82,790 crore) in 2021, a growth of 14 per cent, per a 2022 report by Praxis Global Alliance, a global management consulting and advisory services firm. The report also estimates specialised tests to grow 15-22 per cent by volume and 40-45 per cent by value each year. And while genetic diagnostics in India was valued at $60.78 million (Rs 503 crore) in 2022, the figure is anticipated to grow at a CAGR of nearly 7 per cent, according to a 2022 report by the domain biggie Research and Markets.



WHAT IS PREDICTIVE HEALTHCARE?

Predictive analytics in healthcare entails the use of data to assess the possibility of disease before it has actually set in. It can be used for healthy adults who, if found predisposed to a particular disease, can with the help of lifestyle or medical interventions avert it altogether. In fact, predictive diagnostics can be applied even before a person is born to foretell genetic defects, if any. Which is what 25-year-old Rubina Sengupta, a homemaker from Hyderabad, opted for before planning her pregnancy with her engineer husband after she read a media report on genetic carrier genes in parents and their impact on a child's development and health. She took the Claria Carrier Screening Test, a reproductive genetic screening offered by MedGenome, a nationwide genetic diagnostics chain, to check for mutations in 2,000 genes of would-be parents and predict if they can pass the illness to their yet-unborn child. "If we have certain carrier genes," says Sengupta, "then I feel we should rather adopt than have a biological child."

> "By identifying the exact triggers responsible for allergic reactions, **tailor-made treatment plans** and personalised recommendations can now be offered to patients"
>
> **–DR ARJUN DANG**
> *CEO, Dr Dangs Lab*

Likewise, 46-year-old Shalini Pahwa, a graphic artist from Bengaluru, took a genetic test for endometrial cancer because her mother had it. "They found a mutation in my gene that would cause it and was a guaranteed likelihood," she says. Pahwa, therefore, decided to have her uterus removed right away. "I saw no point in keeping it as the organ would have had to be taken out anyway," she adds.

Predictive analytics has indeed brought about a transformation in cancer detection. Breast cancer, in fact, is one area in which prediction is stronger than ever before. Studies have come up with at least four gene variations, the presence of which can confirm a 100 per cent